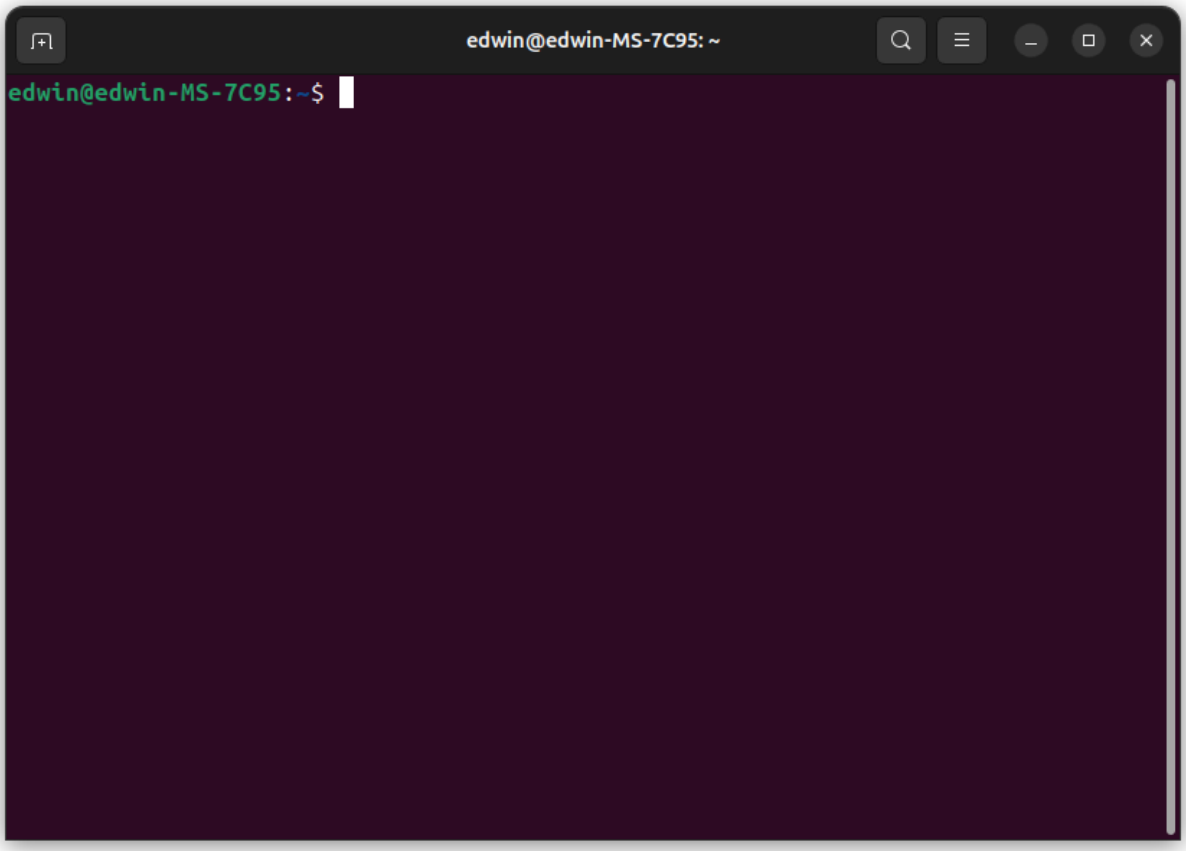
Hardware Needed

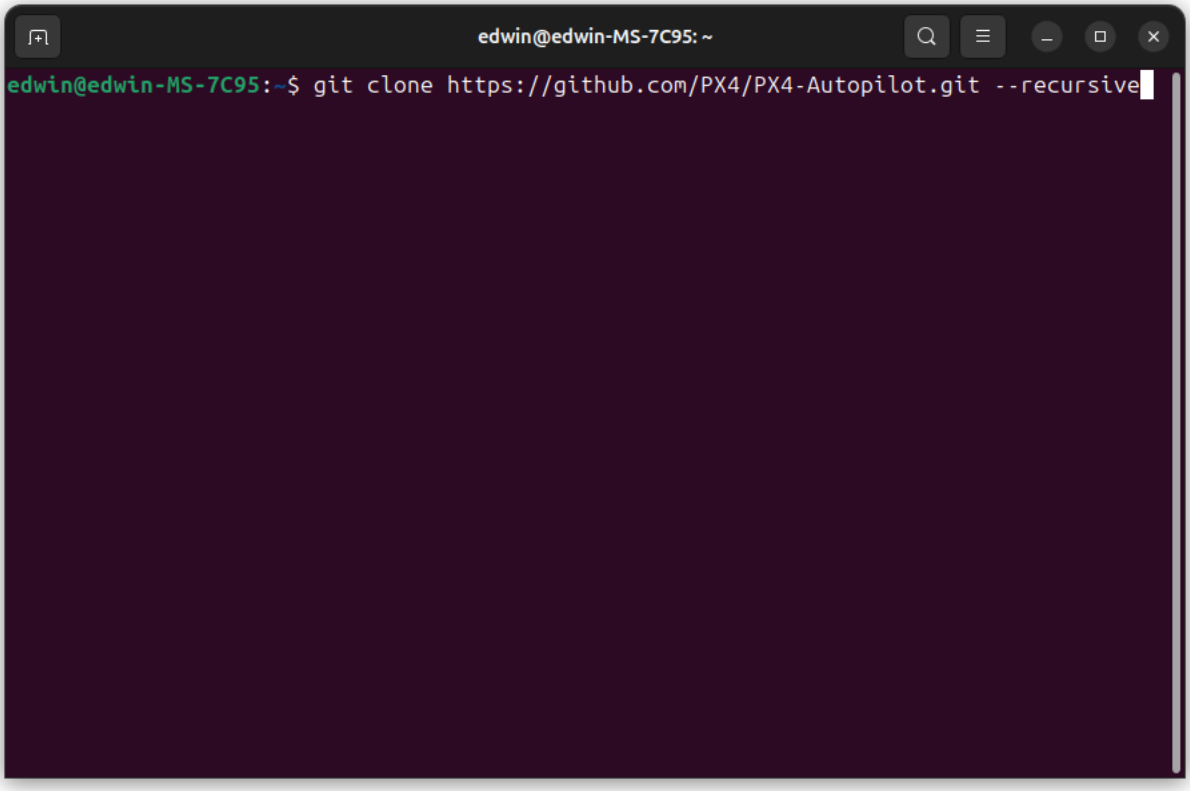
- x86_64 Desktop
    - Ubuntu Linux 24.04
- Raspberry Pi (any model is fine)

Setting up PX4 and Gazebo on Desktop

1. Open the terminal

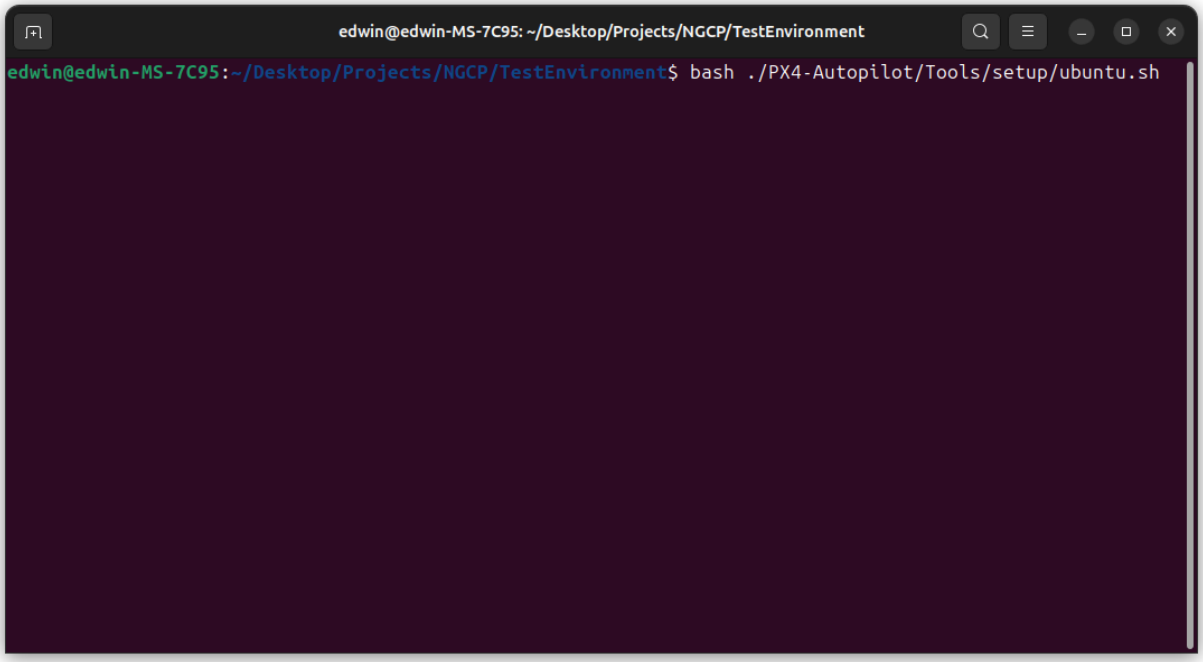


2. Type the following command in the terminal and hit Enter



This command clones the PX4 Autopilot software's GitHub repository to your computer.

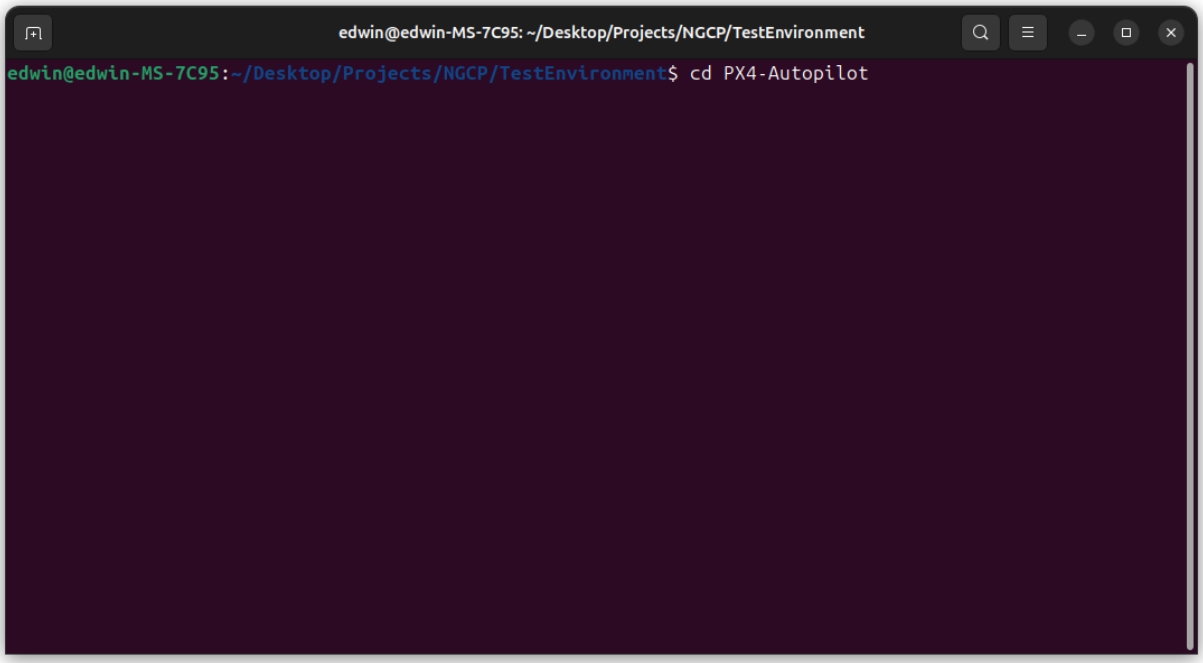
3. Type the following command in the terminal and hit Enter



This command installs the packages required to use the PX4 Autopilot software.

4. Restart your computer after the command has completed installing the necessary packages.

5. Navigate to the folder where you ran the “git clone” command in Step 2. Type this command in the terminal and hit Enter



We must be inside the PX4-Autopilot directory to build our PX4 autopilot software and Gazebo simulation with UAV. Before we do that, however, we need to configure the Mavlink commands that will be executed to set up communication with the Raspberry Pi.

6. Type this command in the terminal and hit Enter.

```
edwin@edwin-MS-7C95: ~/Desktop/Projects/NGCP/TestEnvironment/PX4-Autopilot
edwin@edwin-MS-7C95:~/Desktop/Projects/NGCP/TestEnvironment/PX4-Autopilot$ nano ROMFS/px4fmu_common/init.d-posix/px4-rc.mavlink
```

I'm using nano to edit this file, but you may use any text editor. Assuming you're using nano, you should see something like this:

```
edwin@edwin-MS-7C95: ~/Desktop/Projects/NGCP/TestEnvironment/PX4-Autopilot
  GNU nano 7.2                    ROMFS/px4fmu_common/init.d-posix/px4-rc.mavlink *
#!/bin/sh
# shellcheck disable=SC2154

udp_offboard_port_local=$((14580+px4_instance))
udp_offboard_port_remote=$((14540+px4_instance))
[ "$px4_instance" -gt 9 ] && udp_offboard_port_remote=14549 # use the same ports for more than 10 instances to avoid port overl>
udp_onboard_payload_port_local=$((14280+px4_instance))
udp_onboard_payload_port_remote=$((14030+px4_instance))
udp_onboard_gimbal_port_local=$((13030+px4_instance))
udp_onboard_gimbal_port_remote=$((13280+px4_instance))
udp_gcs_port_local=$((18570+px4_instance))

# GCS link
mavlink start -p -x -u $udp_gcs_port_local -r 4000000 -f
mavlink stream -r 50 -s POSITION_TARGET_LOCAL_NED -u $udp_gcs_port_local
mavlink stream -r 50 -s LOCAL_POSITION_NED -u $udp_gcs_port_local
mavlink stream -r 50 -s GLOBAL_POSITION_INT -u $udp_gcs_port_local
mavlink stream -r 50 -s ATTITUDE -u $udp_gcs_port_local
mavlink stream -r 50 -s ATTITUDE_QUATERNION -u $udp_gcs_port_local
mavlink stream -r 50 -s ATTITUDE_TARGET -u $udp_gcs_port_local
                                          [ Read 41 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line  M-E Redo     M-6 Copy
```

If you don't see anything, exit using Ctrl + x and make sure you typed in the command correctly.

7. Go to the line starting with "mavlink start" that is under the line, "#API/Offboard link"

```
# API/Offboard link
mavlink start -x -u $udp_offboard_port_local -r 4000000 -f -m onboard -o $udp_offboard_port_remote
```

8. Change that line to the following

```
# API/Offboard link
mavlink start -p -x -u $udp_offboard_port_local -r 4000000 -f -m onboard -o $udp_offboard_port_remote -t <IP address of rPi here>
```

Replace “<IP address of rPi here>” with the IP address of your Raspberry Pi.

We add two things to this line: -p and -t <IP address of rPi here>. The -p argument tells Mavlink to broadcast its message to the network, instead of the message only being able to be sent to another application on your computer. The -t argument specifies what device Mavlink should broadcast its message to (in this case, the Raspberry Pi).

Press Ctrl + o, Press Enter to save your changes, and Press Ctrl + x to exit the nano editor.

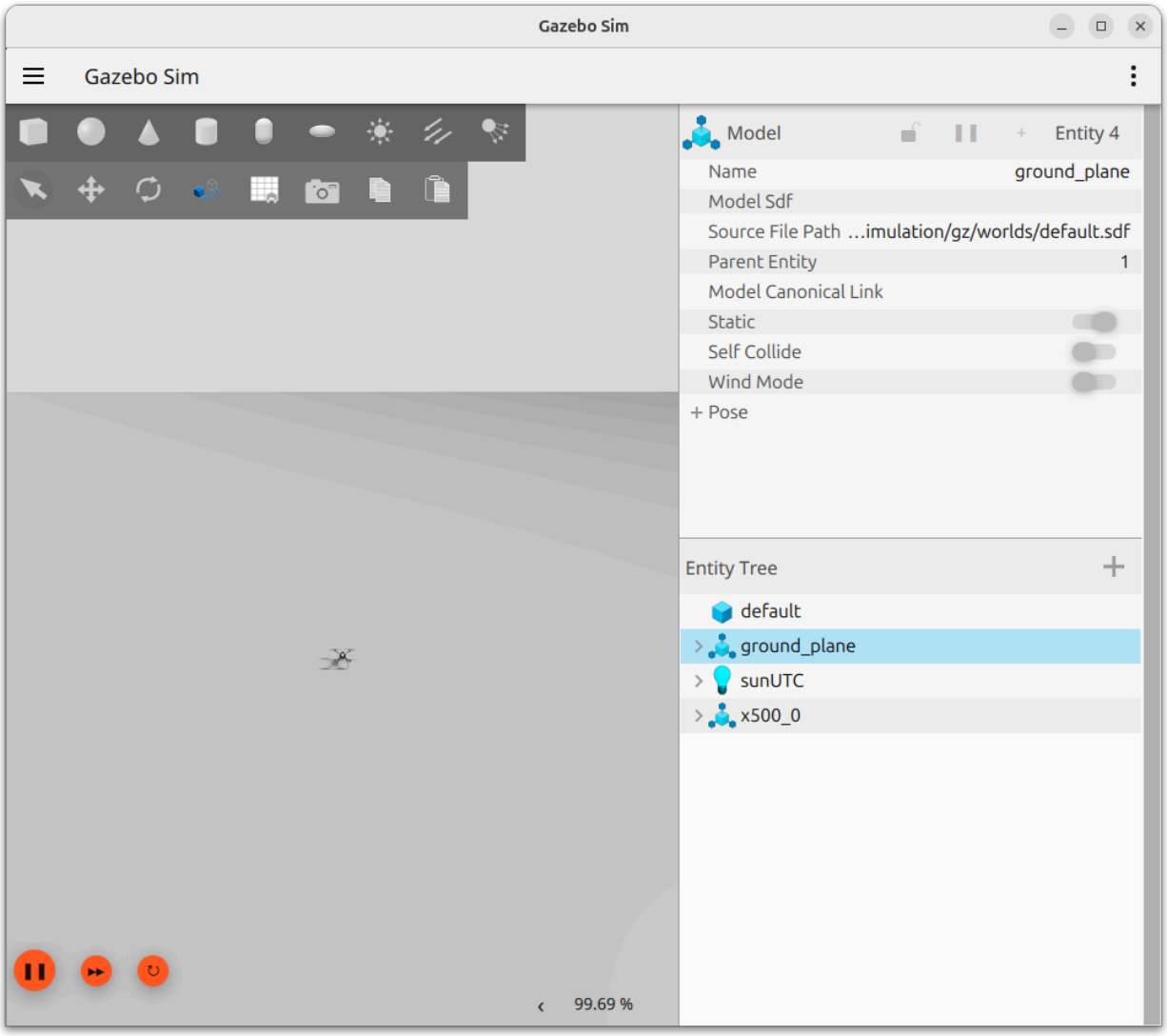
9. Type this command in the terminal and hit Enter

```
edwin@edwin-MS-7C95: ~/Desktop/Projects/NGCP/TestEnvironment/PX4-Autopilot
edwin@edwin-MS-7C95:~/Desktop/Projects/NGCP/TestEnvironment/PX4-Autopilot$ make px4_sitl gz_rc_cessna
```

This will start the PX4 Autopilot software and Gazebo with a fixed-wing UAV. You should see something like this on your terminal:

```
edwin@edwin-MS-7C95: ~/Desktop/Projects/NGCP/TestEnvironment/PX4-...
INFO  [tone_alarm] home set
INFO  [uxrce_dds_client] init UDP agent IP:127.0.0.1, port:8888
INFO  [mavlink] mode: Normal, data rate: 4000000 B/s on udp port 18570 remote po
rt 14550
INFO  [mavlink] using network interface wlp5s0, IP: 192.168.1.129
INFO  [mavlink] with netmask: 255.255.255.0
INFO  [mavlink] and broadcast IP: 192.168.1.255
INFO  [mavlink] mode: Onboard, data rate: 4000000 B/s on udp port 14580 remote p
ort 14540
INFO  [mavlink] mode: Onboard, data rate: 4000 B/s on udp port 14280 remote port
 14030
INFO  [mavlink] mode: Gimbal, data rate: 400000 B/s on udp port 13030 remote por
t 13280
INFO  [mavlink] using network interface wlp5s0, IP: 192.168.1.129
INFO  [mavlink] with netmask: 255.255.255.0
INFO  [mavlink] and broadcast IP: 192.168.1.255
INFO  [logger] logger started (mode=all)
INFO  [logger] Start file log (type: full)
INFO  [logger] [logger] ./log/2025-09-17/04_13_57.ulg
INFO  [logger] Opened full log file: ./log/2025-09-17/04_13_57.ulg
INFO  [px4] Startup script returned successfully
pxh> WARN  [health_and_arming_checks] Preflight Fail: No connection to the GCS
WARN  [health_and_arming_checks] Preflight Fail: No connection to the GCS
```

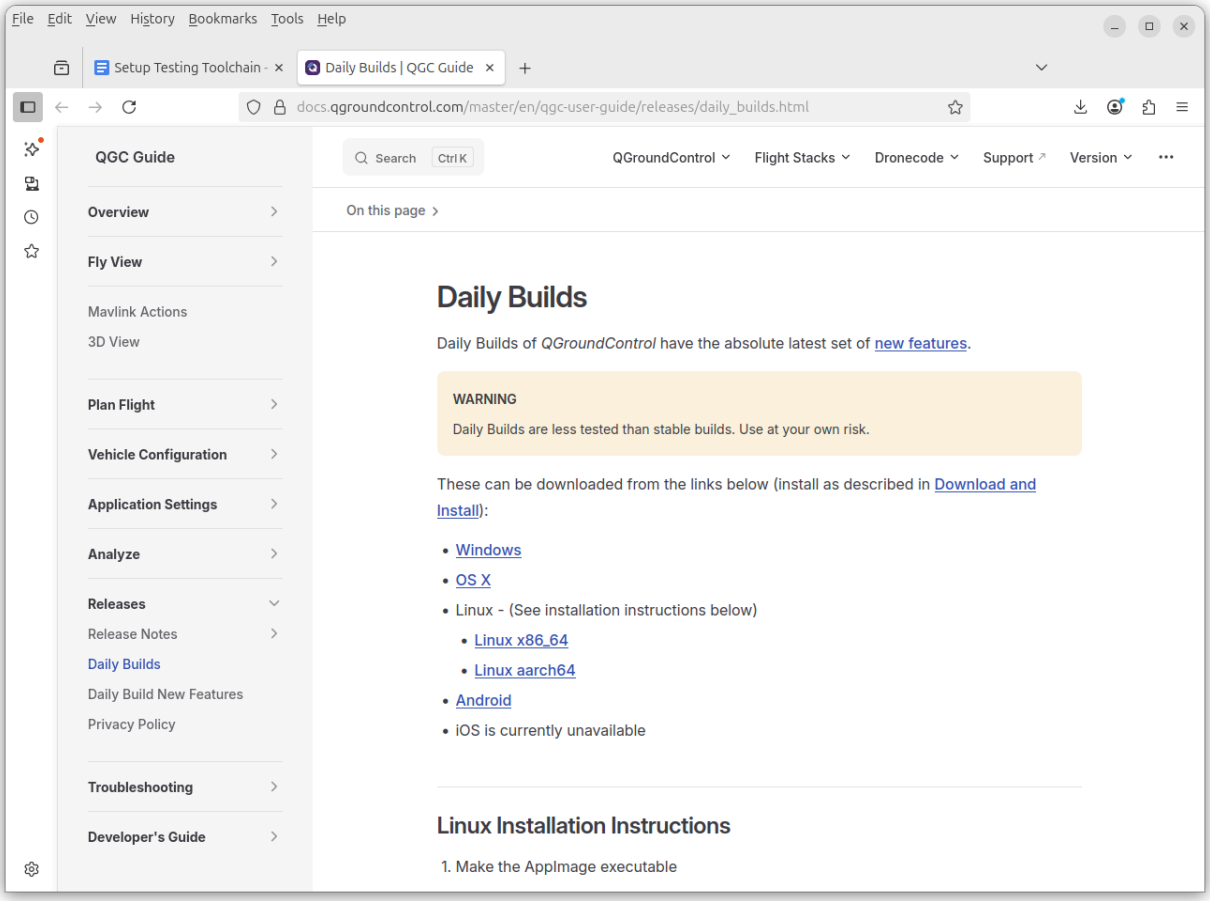
You should also see the Gazebo Simulator running.



This concludes the installation for PX4 and Gazebo on Desktop.

Setting up QGroundControl on Desktop

1. Go to this link. If the link is corrupted, look up “QGroundControl Daily Build” on Google and it will be the first link from the QGroundControl Documentation. You should see something like this.



Click on the “Linux x86_64” link to begin downloading QGroundControl for Linux.

Linux Installation Instructions

1. Make the AppImage executable

```
chmod +x QGroundControl-<arch>.AppImage
```

2. Enable serial-port access Add your user to the dialout group so you can talk to USB devices without root:

```
sudo usermod -aG dialout "$(id -un)"
```

INFO

At login, your shell takes a snapshot of your user and group memberships. Because you just changed groups, you need a fresh login shell to pick up "dialout" access. Logging out and back in reloads that snapshot, so you get the new permissions.

3. (Optional) Disable ModemManager On some Ubuntu-based systems, ModemManager can claim serial ports that QGC needs. If you don't use it elsewhere, mask or remove it.

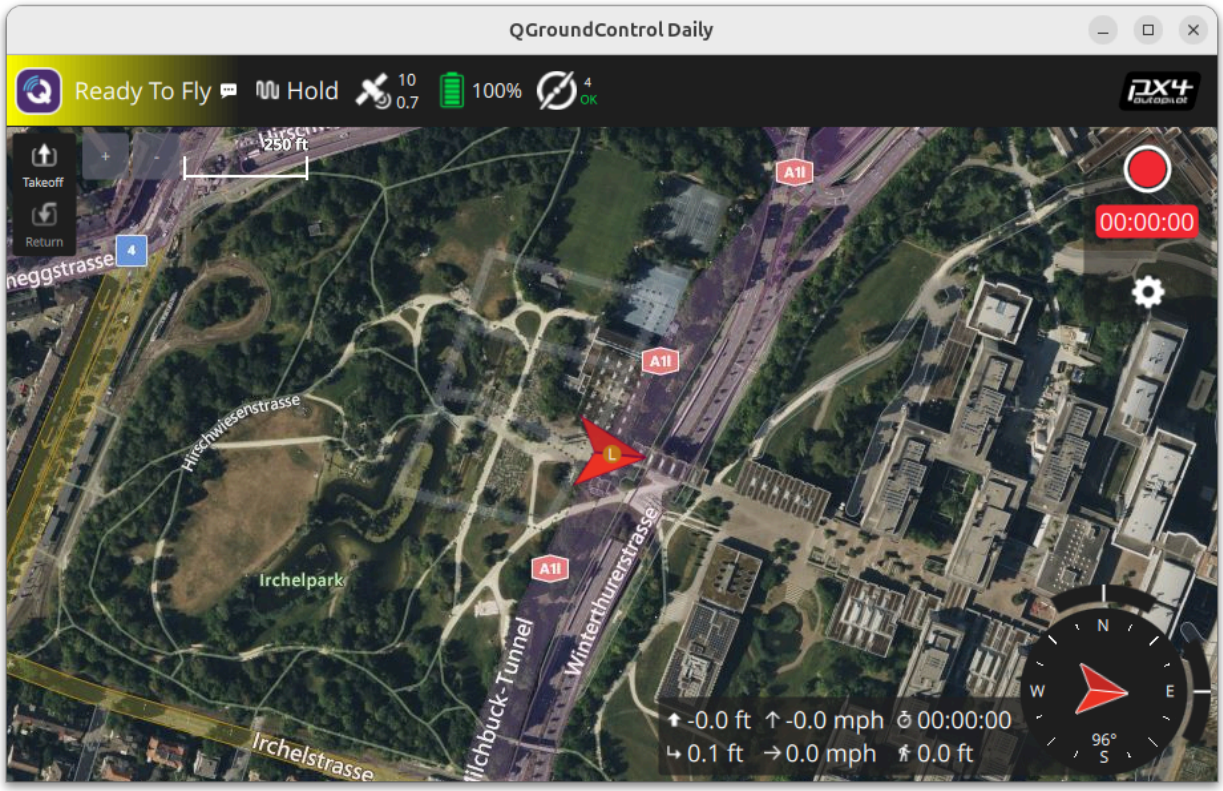
```
# preferred: stop and mask the service
sudo systemctl mask --now ModemManager.service

# or, if you'd rather remove the package
sudo apt remove --purge modemmanager
```

4. Run QGroundControl Either double-click the AppImage in your file manager or launch it from a terminal:

```
./QGroundControl-<arch>.AppImage
```

After completing Step 4 QGroundControl's Linux Install Instructions, QGroundControl will automatically connect with the PX4 Autopilot software. You should see something like this in the QGroundControl application:



This concludes Setting up QGroundControl on Desktop

Setting up MAVSDK Python on Raspberry Pi

NOTE: Screenshots aren't shown because I can't screenshot the Raspberry Pi terminal

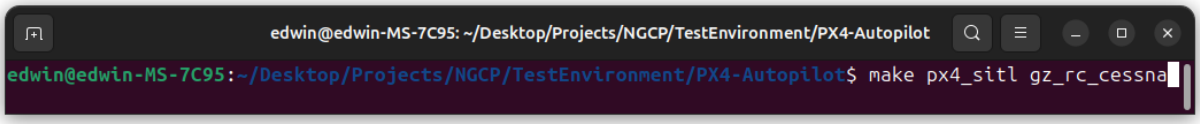
1. Type "pip3 install mavsdk –break-system-packages" to the Raspberry Pi terminal and hit Enter.
2. [Download this file from our GitHub]
3. Run the file using Python3

If you have PX4 Autopilot software and Gazebo running in the background, you will see the drone takeoff, stay in the air for 10 seconds, then land.

This concludes Setting up MAVSDK Python on Raspberry Pi

Post-Installation

1. On the desktop, run PX4 Autopilot software and Gazebo by navigating to the “PX4-Autopilot” directory on your computer on your terminal. Type the following command and press Enter.



2. On the desktop, run QGroundControl by navigating to the directory you downloaded its App Image file and follow Step 4 on QGroundControl Documentation’s “Linux Installation Instructions”.
3. On the Raspberry Pi, run MAVSDK by running the Python file downloaded from our GitHub using python3. You may edit the file to control the plane aside from takeoff and landing.